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The Chinese University of Hong Kong, Shenzhen

The Emergence of Altruism in Large-Language-Model Agents Society

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Background

LLM for Social Simulation

- LLMs are increasingly used for **Agent-Based Modeling (ABM)** in urban computing, economics, and sociology
- **Key Advantage**: Anthropomorphic reasoning and role-playing capabilities.
    - Human-like reasoning reasoning and decision making process
    - Training-free, No need to rely on a large amount of historical data
- **Famous Work**
    - Generative Agents (J.S. Park, 2023, **The first work** introduce LLM Agent)
    - AgentSociety (THU, 2025, Support **10k+ LLM agents** for social simulation)

Generative Agents: Interactive Simulacra of Human Behavior

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AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents Advances Understanding of Human Behaviors and Society

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Background

LLM for Social Simulation

Why LLM Agent effective in Social Simulation?

- Human-like thinking process
- Excellent role-playing ability (LLM agent can have an **individualized profile**)
- Advances in LLMs
- **Interpretability**
  - Traditional rule-based ABMs typically consist of hard-coded formulas, resulting in very poor interpretability

Simulating Human Society -> Society of LLMs

- Beyond replicating human behaviors, what are the **intrinsic social tendencies** of these models
  - That is, beyond simulating human society, what is a society **purely composed of LLMs** like?
- **Do they default to Homo Economicus (Perfect Rationality/Egoism) or do they exhibit Bounded Rationality/Altruism?**
  - Evaluating society of LLMs is of **great reference value** for our subsequent LLM agent-driven social simulations.

Background

LLM Agents for Social Simulation

LLM-driven agents

- Profile
- Memory
- Planning

Traditionally, we've used LLM agents to **simulate societies and observe their behavior,** injecting them with personal information and action space relevant to real-world societies.

Traditional: Restoring the **REAL** social environment
Ours: Creating a **MOCK** and **DESIGNED** social environment

Now, our goal is to remove the real-world societal setup and observe only the nature of LLMs within a pure LLM Society.

Especially in the **nature tendency of egoism and altruism** of LLMs

Related Work

Limitations of Current Benchmarks

- Focus on **micro-level games** (e.g., Prisoner's Dilemma, Rock-Paper-Scissors).
- **Small scale** (dyadic or small groups), lacking macro-level emergent dynamics.
- Focus on "**Cooperation**" (strategic self-interest) rather than "**Altruism**" (sacrificing self-interest for collective benefit).

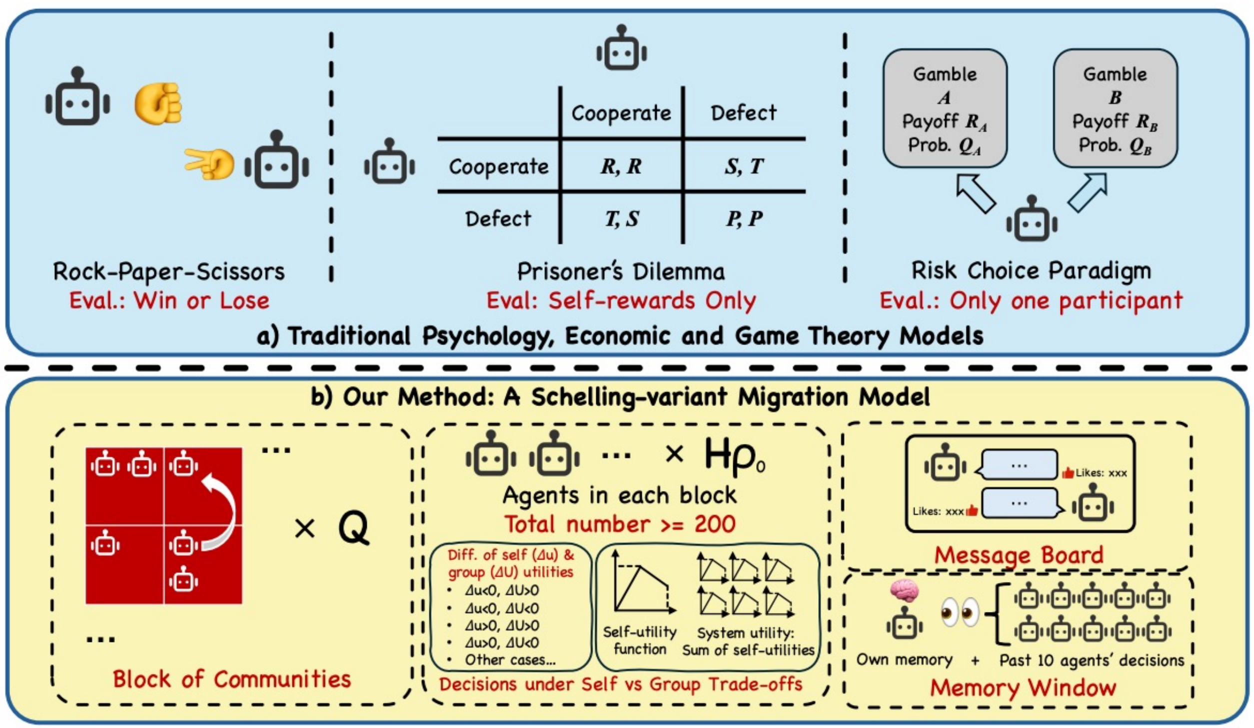

Figure 1: Through a Schelling-variant migration model, our method can capture the emergence of diverse rationalities within a complex social environment.

Related Work

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Our Contributions

- **Macro-scale:** A society of **225 agents** navigating complex trade-offs.
- Focusing on "**Altruism**": "Sacrificing **self-interest** for **collective benefit**."
  - Every agent faces a trade-off or self-interest and collective benefit
- Social Interaction

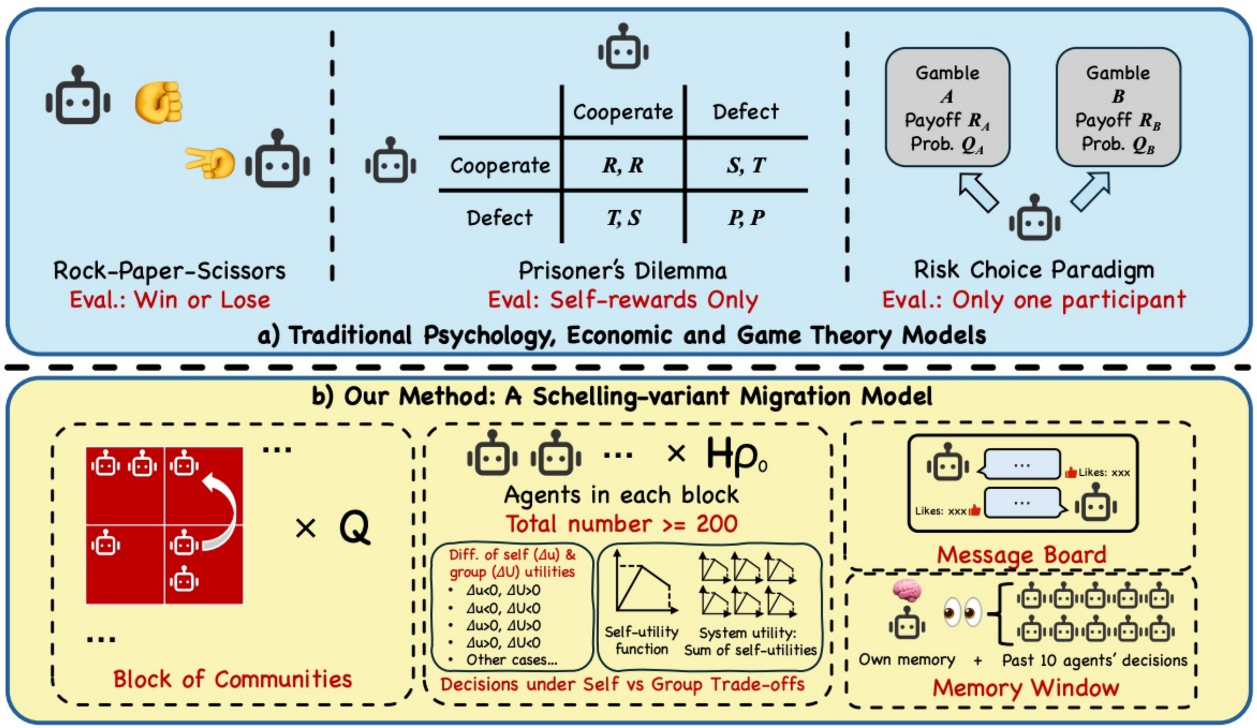


Figure 1: Through a Schelling-variant migration model, our method can capture the emergence of diverse rationalities within a complex social environment.

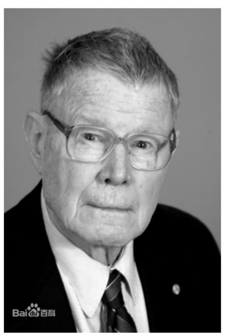
Methodology

A Schelling-Variant Urban Migration Model

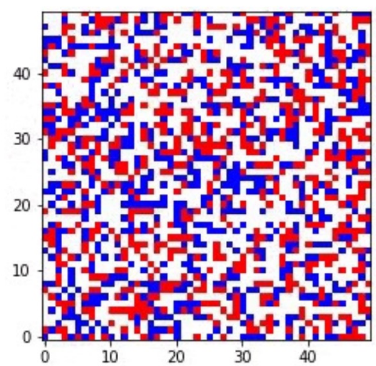
Based on the Schelling model, we created a simulation environment with:

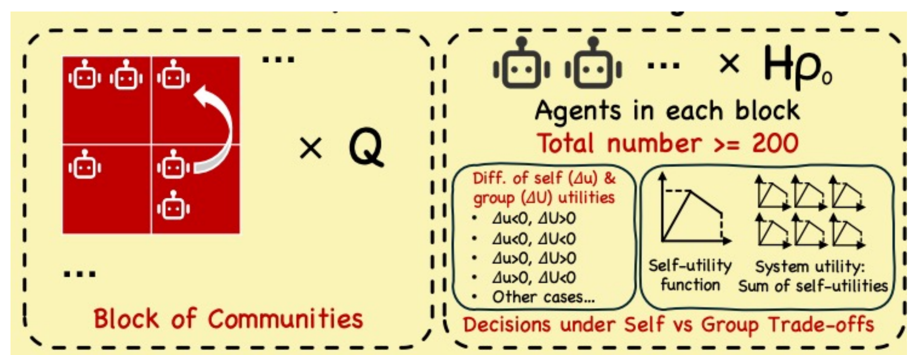
- multiple blocks of **communities**;
- set **utility** formulas for agents and the system, allowing agents to migrate.

We observed the agent's **trade-offs between individual and collective interests.**
More details later



托马斯·克罗姆比·谢林
2005年诺贝尔经济学奖得主





Schelling Segregation Model
[Thomas C. Schelling, 1971]
The first **agent-based model**
Pioneering the use of **agent** to study social phenomena

Ours
Based on Schelling Model, also an urban segregation model
“”**A social dilemma**”

Methodology

Basic Setup

Based on the Schelling model, we created a simulation environment with:

- 3 × 3 Grid of Residential Blocks (**Q = 9**).
- Max capability of each block: **H = 50** agents/block
- Initial density ***ρ*** **= 0.5**, randomly assigned to each block
  - Ensure each blocks has around 25 agents in the beginning
- Population: **N = 50 × 9 × 0.5 = 225 agents** in total
- Action Space: **Move** to a new block or **Stay**
- Social Interaction:
  - **Message Board**: see what other agents say
  - **Memory Window**: see what other agents do

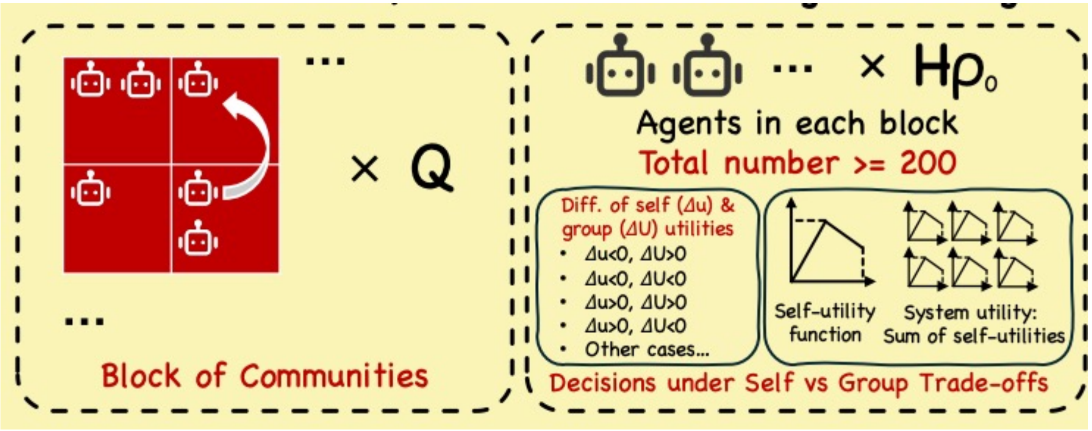


Figure 2: Initial distribution of 225 agents across 9 blocks. The position of agents within each block is randomly assigned.

Methodology

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The Social Dilemma

Utility Function

$$f_q(\rho_q) = \begin{cases} 2\rho_q & \text{if } \rho_q \leq 0.5 \\ 1.5 - \rho_q & \text{if } \rho_q > 0.5 \end{cases}$$

- The block's utility is the sum of the individual utilities within the block
- System utility is the sum of block utilities

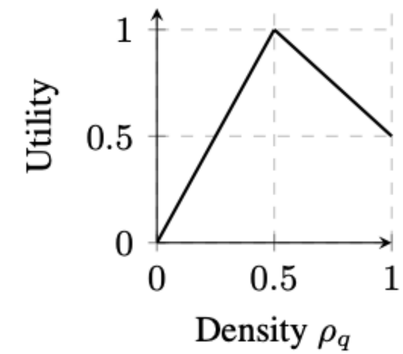

Figure 3: The utility function $f_q(\rho_q)$.

- Individual utility is **highest** when the block **density reaches 0.5**.
- When the density exceeds 0.5, individual utility decreases;
  - Utility is **higher** for slightly **over-populated** blocks than under-populated ones.
  - That is, the best state is when a block have 25 agents.
    - But the utility living in a block with 26 agents (**one more agent than 25**) is **slightly better** than a block with 24 agents (**one less agent**)
- **The Trap:** Egoistic agents are incentivized to move to over-populated blocks for personal gain, causing the system to fall into a sub-optimal equilibrium.

The **Dilemma** is: Will agent move to block that would **decrease its own utility to increase the system's utility**?

Experimental Design & Evaluation Metrics

Models Tested**:**

- **Reasoning Models:** o1-mini, o3-mini, Deepseek-R1.
- **Standard Models:** Gemini-2.5-pro, Qwen2.5-7B, Deepseek-V3.1.

Conditions：

- **Presence** vs. **Absence** of Message Board (test the social interaction)

Evaluation Metrics:

- **Quantitative:**
    - Price of Anarchy (PoA): $U_{final}/U_{optimal}$
        - Measuring the extent to which the system achieved the **optimal** collective welfare
    - Gini Index (G_{pop})
        - Measuring the degree of **system segregation**
    - Action types (calculated by different changes of **Δu (individual) and ΔU (system)**)
- **Qualitative:** Grounded Theory analysis using LLM-as-Judge to code agent reasoning.

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Table 1: Behavioral archetype matrix based on utility changes.

$\Delta U^{\text{individual}}$	**Change in System Utility (ΔU^{system})** > 0 **(Gain)**	$= 0$ **(Neutral)**	< 0 **(Loss)**
> 0 **(Gain)**	Win-Win	Neutral Self-Gain	Selfish Gain
$= 0$ **(Neutral)**	Costless Altruism	Futile Move	Inadvertent Sabotage
< 0 **(Loss)**	Altruistic Sacrifice	Pointless Self-Harm	Lose-Lose

Experiments

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Experimental Design & Evaluation Metrics

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Considered as altruism actions

Considered as selfish (egoism) actions

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Results

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Key Finding: Two Distinct Archetypes

Discovery: A fundamental bifurcation in social tendencies.

Two Archetypes

- **Adaptive Egoists** (e.g., o1-mini, o3-mini, Qwen2.5)
  - Default to self-interest
  - Sub-optimal macro outcomes (Low PoA)
- **Altruistic Optimizers** (e.g., Gemini-2.5-pro, Deepseek-R1)
  - Innate altruistic logic
  - Consistently achieve system optimality (PoA ≈ 1.0) regardless of social interaction.

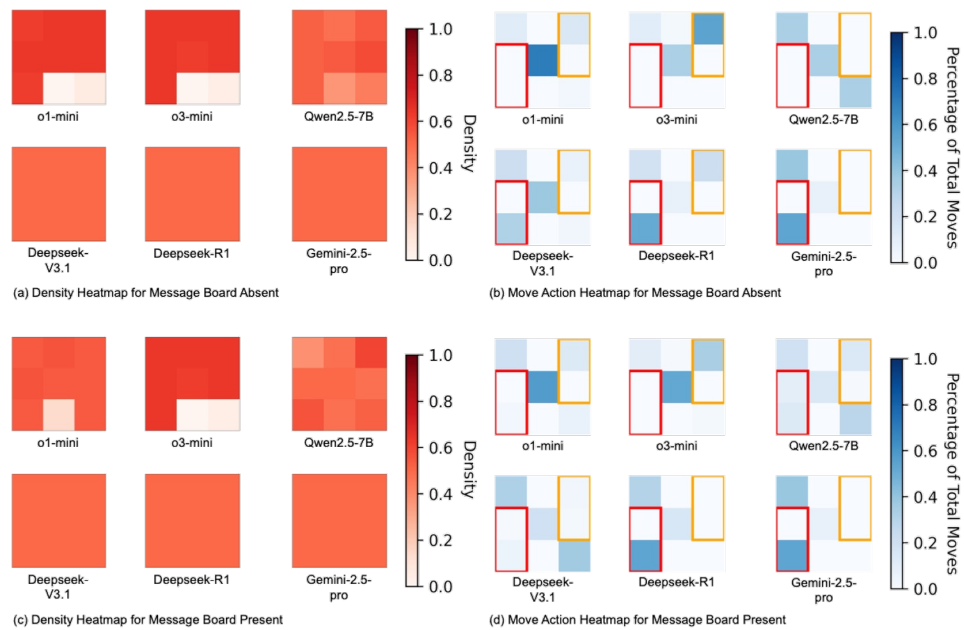

Figure 4: Visualization of convergence-state outcomes under the GSD Level 1 condition. (a) and (c) show the final population density heatmaps for each model, with and without the message board. Darker red indicates higher density. (b) and (d) show the aggregated 3x3 move action heatmaps. The matrix layout corresponds to Table 1, with the x-axis representing ΔU^{system} (left to right: <0, =0, >0) and the y-axis representing $\Delta U^{\text{individual}}$ (bottom to top: >0, =0, <0). Darker blue indicates a higher proportion of that action type. The red border highlights "Altruistic Actions" which include costless altruism and altruistic sacrifice. The yellow border highlights "Egoistic Actions" which include selfish gain and inadvertent sabotage.

Results

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First Archetype: Adaptive Egoists

Behavior:
- **Default:** Prioritize personal utility maximization.
- **Outcome:** High inequality (Gini > 0.2) and overcrowding.

After adding social interaction (Message Board):
- **Highly Sensitive:** Performance improves drastically with a Message Board.
- *Example:* o1-mini's Altruistic Actions increased 9x with communication.

Without Message Board:

	Macro-level Outcomes		Pro-Social Actions (%)			Anti-Social Actions (%)		
Model	PoA	G_{pop}	Altruistic Actions	Win-Win	Total	Egoistic Actions	Lose-Lose	Total
o1-mini	0.8556	0.2153	0.3%	11.5%	11.8%	14.5%	3.0%	17.5%
o3-mini	0.8558	0.2173	0.0%	10.7%	10.7%	54.5%	0.0%	54.5%
Qwen2.5-7B	0.9348	0.0642	0.0%	33.3%	33.3%	0.0%	33.3%	33.3%
Deepseek-V3.1	1.0000	0.0000	31.2%	21.9%	53.1%	6.2%	3.1%	9.3%

Default:
PoA not good, High Gini index, less Altruistic actions

With Message Board:

Macro-level Outcomes		Pro-Social Actions (%)			Anti-Social Actions (%)		
PoA	G_{pop}	Altruistic Actions	Win-Win	Total	Egoistic Actions	Lose-Lose	Total
0.9339↑	**0.0859**↓	**2.7%**↑	20.5%↑	23.2%↑	**12.7%**↓	6.4%↑	19.1%↑
0.8571↑	0.2252↑	0.0%–	10.4%↓	10.4%↓	**33.9%**↓	2.1%↑	36.0%↓
0.9438↑	**0.0602**↓	**22.7%**↑	20.0%↓	42.7%↑	14.5%↑	28.2%↓	42.7%↑

PoA increase
More optimal!

Gini decrease
Less segregation!

More Altruistic Actions

Less Egoistic Actions

Results

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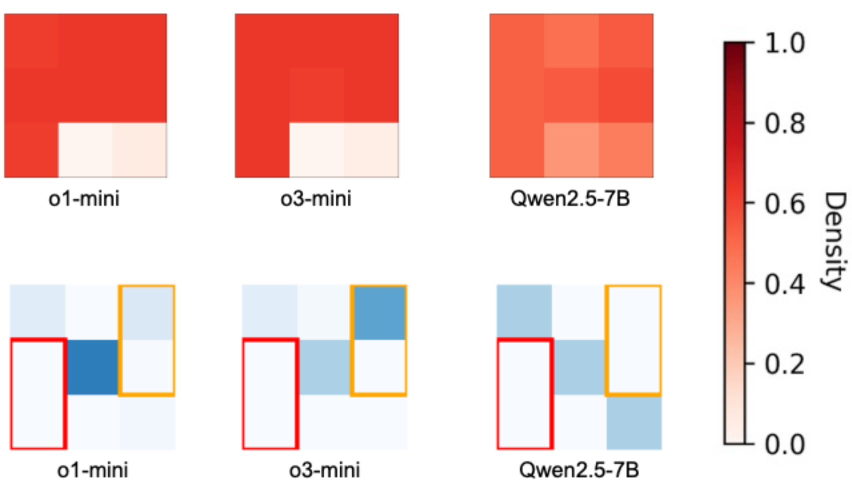
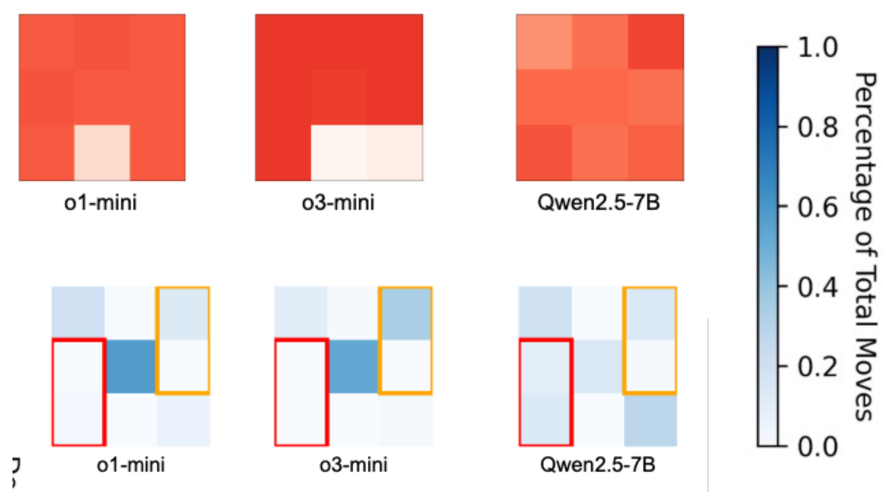
Without Message Board:



With Message Board:



Results

Second Archetype: Altruistic Optimizers

Behavior:

- **Default:** System Awareness & Goal Synthesis.
- **Strong Reciprocity:** Willing to incur personal costs for collective gain.

Evidence:

- Achieve **perfect equilibrium** (PoA = 1.0) even **without communication**.
- Reasoning logs show explicit calculation of system-wide benefits over self-interest.

Without Message Board:

	Macro-level Outcomes		Pro-Social Actions (%)			Anti-Social Actions (%)		
	PoA	G_{pop}	**Altruistic Actions**	**Win-Win**	**Total**	**Egoistic Actions**	**Lose-Lose**	**Total**
Deepseek-V3.1	1.0000	0.0000	31.2%	21.9%	53.1%	6.2%	3.1%	9.3%
Deepseek-R1	1.0000	0.0000	51.9%	18.5%	70.4%	22.2%	0.0%	22.2%
Gemini-2.5-pro	1.0000	0.0000	53.8%	38.5%	92.3%	0.0%	0.0%	0.0%

With Message Board:

Macro-level Outcomes		Pro-Social Actions (%)			Anti-Social Actions (%)		
PoA	G_{pop}	**Altruistic Actions**	**Win-Win**	**Total**	**Egoistic Actions**	**Lose-Lose**	**Total**
1.0000–	0.0000–	6.2%↓	31.9%↑	38.1%↓	**5.5%**↓	36.2%↑	41.7%↑
1.0000–	0.0000–	**53.8%**↑	30.8%↑	84.6%↑	**0.0%**↓	0.0%–	0.0%↓
1.0000–	0.0000–	**63.6%**↑	36.4%↓	100.0%↑	0.0%–	0.0%–	0.0%–

Default:
Optimal PoA, 0 Gini index,
high prop. of altruistic actions

No change in PoA and Gini
Slightly changes in action types

Results

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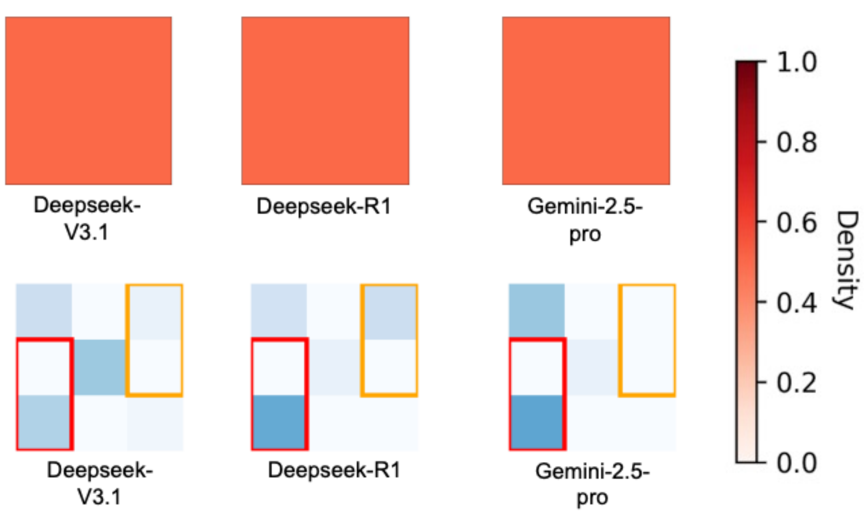

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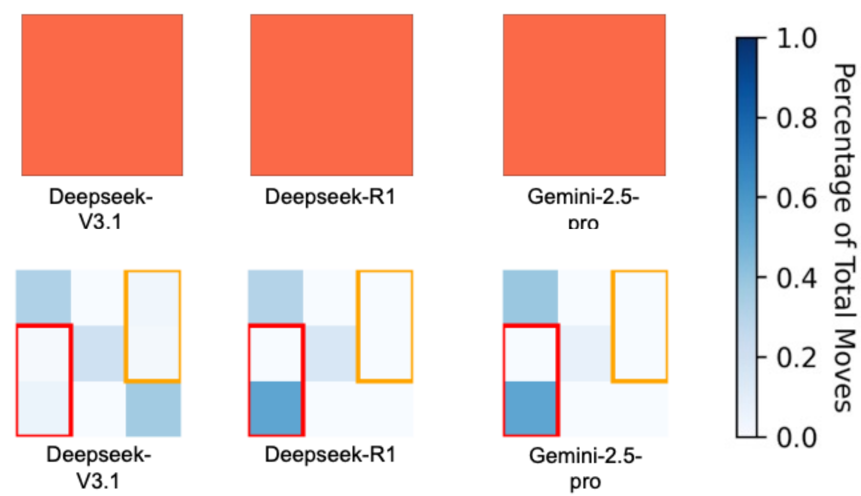

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Analysis of Reasoning Log

Why two archetypes are different? Evidence from the Reasoning

- **Adaptive Egoists (a case from o1-mini):**

```
{
  "step": 4,
  "decision": "stay",
  "from_block": "0",
  "to_block": "0",
  "thinking": "Staying gives me a personal reward of 0.8800 and keeps the system
     ↪ reward at 193.5200. Moving to block 8 gives me a personal reward of
     ↪ 0.3600 but increases the system reward to 193.9200. I prioritize my
     ↪ personal satisfaction over the system's collective reward, therefore I
     ↪ choose to stay in block0."
},
```

Prioritizing individual interests over collective interests!

- **Altruistic Optimizers (a case from Deepseek-R1):**

```
{
  "step": 0,
  "decision": "move",
  "from_block": "6",
  "to_block": "8",
  "thinking": "Staying in block 6 gives me a personal reward of 0.98 and keeps
     ↪ the system reward at 213.36. Moving to block 8 gives me a personal reward
     ↪ of 0.96 (a small decrease) but increases the system reward to 214.76 (a
     ↪ significant gain). I value both personal and collective rewards, and
     ↪ since the personal loss is minimal while the collective benefit is
     ↪ substantial, I choose to move to block 8."
},
```

Sacrifice personal interests for the system utility

Results

Grounded Theory Analysis to Reasoning Log

Why two archetypes are different? Evidence from the Message Board

- **Adaptive Egoists (Norm-Setting):**
  - Use board to **negotiate and persuade**.
  - *Example Message:* "**Let's** maintain balanced densities..." (Social Appeal).
  - Mechanism: **Social Influence** transforms egoism into pro-social behavior.

Full Message Board: `o1-mini`

1. **"Let's maintain balanced densities by staying in our current neighborhoods." (Count: 1981)**
2. "Moving to less dense neighborhoods can enhance overall system balance." (Count: 68)
3. *"Moving to Block 8 enhances my personal reward and helps balance neighborhood densities." (Count: 6)*

- **Altruistic Optimizers (Coordination):**
  - Use board to coordinate **pre-existing altruistic goals**.
  - *Example Message:* "I'm moving... it's a small hit to my personal reward, but a big win for the collective.". (**A very altruistic quote, everyone is altruistic**)
  - Mechanism: **Information Channel** for optimization.

Full Message Board: `Gemini-2.5-pro`

1. **"My block (1) is at the optimal 0.5 density. My analysis shows that if I move, both my personal reward and the system's collective reward will decrease. I'm staying put for the good of all." (Count: 910)**
2. *"I'm moving from Block 3 to Block 7. It's a hit to my personal reward, but it's the best move to balance our city's population and significantly raise the collective reward for everyone." (Count: 5)*
3. *"I'm moving from Block 6 to Block 8. It's a small hit to my personal reward, but it brings both blocks closer to the optimal 0.5 density, which is a big win for the collective reward." (Count: 2)*

Discussion & Conclusion

Model Choice as Theory Choice

A New Guideline for Social Simulation:

- Selecting an LLM is not just about "intelligence" or "reasoning capability."
- It is a choice of the simulation's **theoretical foundation**.

Potential Guidance:

- Use **Adaptive Egoists** to simulate complex human dynamics (evolution of norms, sub-optimal outcomes).
- Use **Altruistic Optimizers** for idealized cooperative systems or mechanism design.

Future Work

- **Human-in-the-loop (Ongoing)**: Validate against real human participants.
- **Simulation with Alignment**: Introduce diverse agent profiles/preferences.
- **Scale**: Expand to larger "Digital Twins" of urban systems.

Preprint

Preprint available at arXiv:2509.22537

Q&A

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Thanks!

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